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2005 Mathematical Contest in Modeling (MCM) Summary Sheet

(Attach a copy of this page to each copy of your solution paper.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

Summary

In this paper, we first propose model 1 to predict the depletion date of the petroleum. In the model, we use the Hubbert curve, an effective curve for predicting oil production, to fit the historic data on oil consumption and discovery. This model can predict the future oil reserve of each year. We make a prediction that the oil will be run out in 57 years.

Using the result of model 1, we develop model 2 by adding the economic, demographic, and political factors into it. The consumption increases with the growth of economy and population, however, the decreasing remnant oil restrains the consumption. This model is based on the relation. From the output of model 2, we find that the economy and population both have significant effect on oil depletion, while the political factors affect the oil consumption apparently in a short-term but influence the oil depletion little. In addition, based on the effect of demand and supply on price, we propose model 3 to predict the future oil price till 2062.

To ensure the oil “security” for the country, we propose to build a Strategic Petroleum Reserve (SPR). We use the economics theory to analyze the cost and benefit of SPR. We put forward two criterions to determine the optimal level of SPR. For the country with much money, the criterion is to maximize the level of SPR with no loss. For the poor country, the criterion is to maximize the retained benefits.

Finally, we put forward some policy to sustain the oil usage over a long time, and compare some alternatives to oil. We also advance some storage and transportation policy to prevent the environment effect of oil, and make a recommendation that the world should extend the use of the renewable energy to get out of the oil crisis.

When will the oil run out

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Introduction

Petroleum is derived from organic material under conditions that were met only rarely in the Earth's long history and then only in a few places. Oil comes primarily from algal material. Such organic debris settled to the floor of the lake or sea in which it lived, and in most cases was oxidized by bottom dwelling organisms or currents.

Nowadays, oil is becoming the most important energy for the world, and it is just like the society's lifeblood. The economy cannot develop without oil. The car cannot move without oil. Some of the Fertilizers and herbicides are oil-based. Oil can also be used to make the plastic products. Oil has entered into all aspects of the people's life.

However, oil is a resource in finite supply; no major oil fields have been found since 1976, and experts suspect that there are no more to find. Some analysts argue that world production is already at or near its peak, although most say that technological progress, which allows the further exploitation of known sources like the Canadian tar sands, will allow output to rise for another decade or two. Therefore, to predict when is the date of depletion is really a crucial question.

Because a little change of the oil price will have a great impact on the world economy, to predict the oil price is also a valuable task. As the amount of oil decrease, the disruption of oil may occur. To protect the oil security, each country should build a Strategic Petroleum Reserve (SPR). The optimal level of SPR needs to be determined.

To delay the depletion year of oil, we need to propose some kinds of policy to incent the oil production and restrain the oil consumption. To get out of the oil crisis, a proper measure such as finding an alternative for its purpose should be take to mitigate the use of oil.

Data Source

Our data comes from the Statistical Review of World Energy published on 15 June 2004. [<http://www.bp.com/subsection.do?categoryId=95&contentId=2006480>]. It is published by BP corporation annually. This publication has been updated with 2003 data and the historic data series has been revised where appropriate. The data we have used are shown in the Appendix A1-A5.

Hubbert Curve

Back in the 1950s a Texas geologist named M. King Hubbert developed the Hubbert curve, which has become the primary analytical tool for understanding and predicting the yield of an oil field, from its discovery to its exploitation, depletion and abandonment. Then he used the curve to predict the peak in US oil production, then world production by analyzing all the oil fields.

[<http://www.gulland.ca/depletion/depletion.htm>]

Peak Oil

Peak oil is where global oil production peaks and, due to continuing demand, prices escalate to a point where current uses are not sustainable.[Ainsley Thomson. 2005]. After the peak oil, the production will decrease, and the supply cannot meet the increasing demand. The price of the oil will escalate significantly. Therefore, peak oil is an important specification for the prediction of oil.

Model 1 Prediction for the Petroleum Depletion

We know the petroleum is a kind of nonrenewable resource, and the total amount of oil extracted over time would follow a logistic curve [Hubbert 1956]. But in practice it is more convenient to use the derivative of logistic curve (called Hubbert curve) to model annual productions [J.H. Laherrère 2000].

The oil produced will be used in two aspects: consumption and storage. Their relation is

$$\text{production} = \text{consumption} + \text{storage}$$

Because of the storage is far less than the consumption (the average percentage of annual storage to annual production between 1965 and 2003 is 0.22%), we consider consumption is approximate the same as production. Hence we can use Hubbert curve to fit annual consumptions function. The general form of Hubbert curve is as follow:

$$c = \frac{2c_m}{1 + \cosh[b_c(t - t_{mc})]} \quad (1)$$

($\cosh$ is the hyperbolic cosine equaling $(e^x + e^{-x})/2$)

where

c is annual consumption;

t is the year;

c_m is production at peak;

t_{mc} is the year of peak;

b_c is a factor describing the slope.

The data we find in website about worldwide petroleum annual consumption is from 1965 to 2003 (Appendix A1). However, in 1970s, OPEC controls the petroleum production effectively, so the consumption becomes abnormal. (see Figure 1). So we only use the annual consumption data from 1985 to 2003.

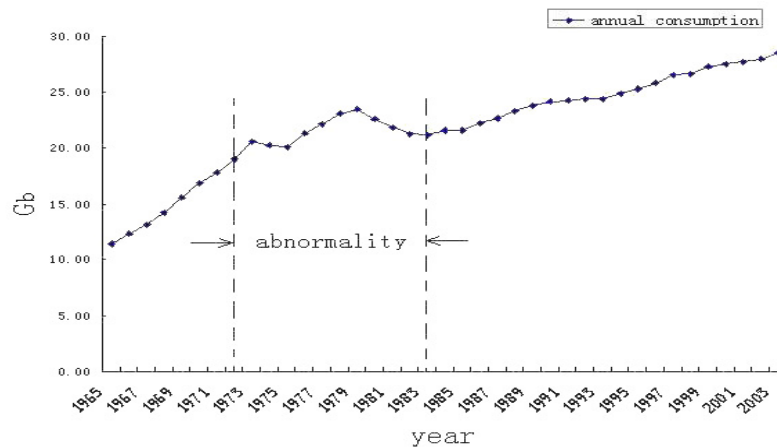


Figure 1 Annual Consumption from 1965 to 2003

Using nonlinear fitting with MATLAB program (see Appendix B1), we find:

$c_m=29.2\text{Gb}$ (thousand million barrels), consumption at peak;

$t_{mc}=2013$, the year of peak;

$b_c=0.0403$, the factor describing the slop.

From the result of fitting, we can see that the peak oil consumption will occur in 2013, and the maximum of annual consumption is about 29.2Gb. After that, the annual consumption will reduce. The annual consumption is shown in Figure 2.

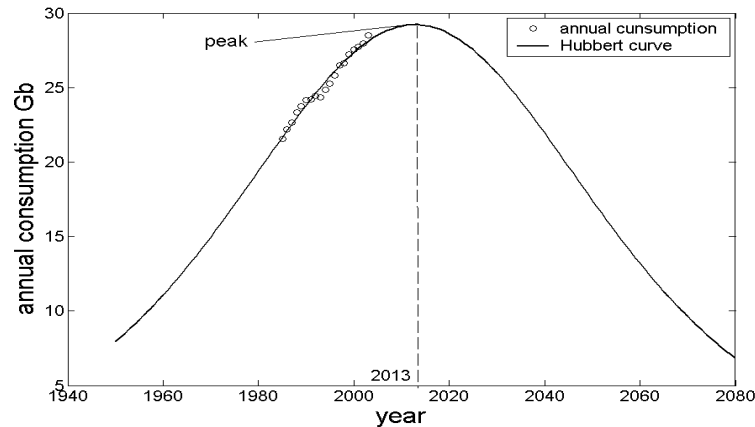


Figure 2 The Result of Fitting for Annual Consumption

Since the natural petroleum reserve is limited, the more oil we discovered, the less is left, and this relation is just like the production of oil. It means the annual discoveries can also be fitted by Hubbert curve as follow:

$$d = \frac{2d_m}{1 + \cosh[b_d(t - t_{md})]} \quad (2)$$

where

d is annual discoveries;

t is the year;

d_m is discoveries at peak;

t_{md} is the year of peak; and

b_d is a factor describing the slope.

The data of annual discoveries from 1980 to 2002 is showed in Figure 3. It can be divided into two segments. One is from 1980 to 1990 except 1985 and 1987. The other is from 1990 to 2002. The discoveries of the former segment have a trend of decreasing with time, while the discoveries of the latter segment have a trend of increasing with time.

The raise of discoveries in the latter segment is possibly caused by the deepwater exploration [J.H. Laherrere. 2000]. Also this slight rise is explained by oil coming on from the former Soviet Union after years of under-production [Aftermath. 1999]. Therefore we cannot use this segment of data because there is still a lot of uncertain factor on deepwater exploration, and we do not have enough data about it.

From Figure 3 we find that the discoveries in 1985 and 1987 are extremely great. The reason could be the fortune factor that causes the abnormal discovery of the petroleum exploration in the two years. Therefore, we put these two data away, and only use the former segment of data to fit the Hubbert Curve.

From Figure 3 we also find that the discoveries in the adjacent years change a lot, and the history discovery data are too few, hence we can foresee these data may not fit the Hubbert Curve exactly.

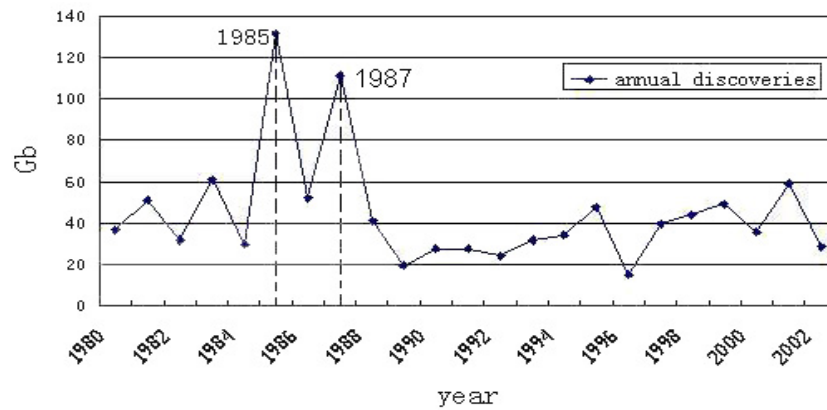


Figure 3 Annual discoveries from 1980 to 2002

Again using nonlinear fitting with MATLAB program (see Appendix B2), we find:
 $d_m = 64.4\text{Gb}$, discoveries at peak;
 $t_{mc} = 1962$, the year of peak; and
 $b_c = 0.0657$, the factor describing the slop.
 The annual discoveries are shown in Figure 4.

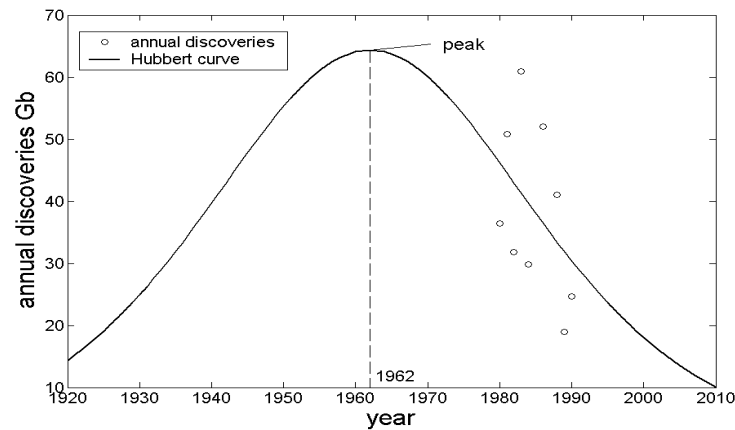


Figure 4 The Result of Fitting for the Annual Discoveries

The worldwide petroleum reserve is related to last year reserve, annual consumption and annual discoveries. The petroleum reserves in $t+1$ year is given by

$$r(t+1) = r(t) + d(t) - c(t) \quad (3)$$

where $r(t)$ is the reserve at t year, $d(t)$ is the discovery at t year, and $c(t)$ is the consumption at t year.

Based on $r(2002) = 1146.3\text{Gb}$ (Appendix A3), equation (1), and equation (2), we can find annual petroleum reserve by equation (3). The depletion year in which the worldwide petroleum reserve decreases to zero is 2062 (see Figure 5). It means the oil will be run out in 57 years. Before the predicted depletion year, there will be 235 Gb oil discovered.

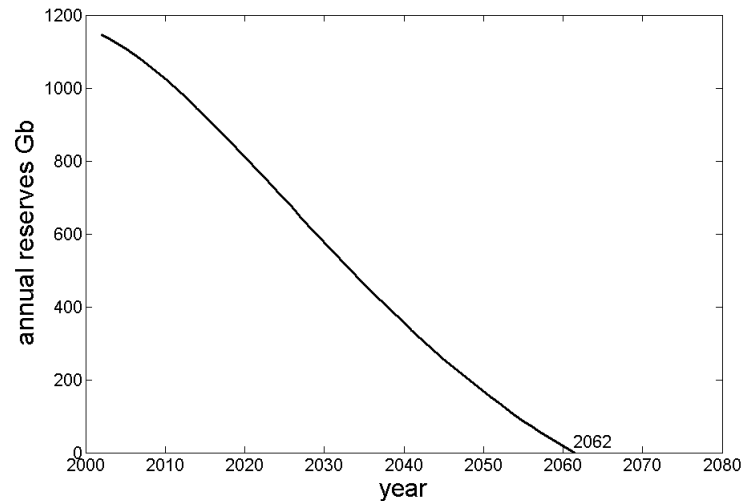


Figure 5 Forecast of Annual Reserve

Model 2 Improved model including economic, demographic and political factors

The model 1 doesn't include the parameters of economic, demographic and political factors. In fact it is a model in which we consider economy and population increase is at the current rate. Moreover, it does not take into account the significant political effects. If the economy and population growth rate is not the same as the current average level, or there is some kind of political factor (such as the OPEC reduce the oil production), the model 1 will lose its effect. Based on the conclusion of the model 1, we develop model 2 that can adapt to more complicated situations.

The development of economy will cause the increase of oil consumption. The number of population has the same impact on oil consumption as economy. We assume that the consumption is in scale with the GDP (Gross Domestic Product) and the population. Because oil is a nonrenewable resource, the remnant oil will decrease with the production. The decreasing remnant oil will limit the increase of consumption. Assuming that the relation between total remnant oil and the consumption is direct ratio, we get the equation:

$$\frac{Qc_d(t)}{g_d(t)p_d(t)r_d(t)} = l$$

where

l is a constant ratio factor

Q is the political factor

$c_d(t)$ is the consumption of oil in t year

$g_d(t)$ is the world GDP in t year

$p_d(t)$ is the population in the world in t year

$r_d(t)$ is the total remnant amount of oil in the world in t year

When there is no political policy that affects the oil consumption, the parameter Q is a constant equaling to 1. When significant political events occur the value of Q will change. If the political factor is larger than 1, it means there is a political event that restrain the consumption. To the contrary, if the political factor is smaller than 1, it means there is a political event that facilitate the consumption. The

more the deviation of the political factor from 1, the greater the impact of the political event on the oil consumption is.

Assuming the GDP grows at average rate of λ and population grows at average rate of μ , we get:

$$g_d(t+1) = g_d(t)(1+\lambda)$$

$$p_d(t+1) = p_d(t)(1+\mu)$$

The remnant amount of oil will decrease as the oil is consumed year by year, we get:

$$r_d(t+1) = r_d(t) - c_d(t)$$

We assume that if the remnant amount of oil in the world is less than current consumption per year, the oil should quit the energy market and the first year less than the current consumption is the depletion year of oil.

From the result of model 1, we get the amount of oil discovered before depletion year is 235Gb, and in the reserve data sheet (Appendix A3), we know the oil reserve in 2003 is 1147Gb, so we have $r_d(2003) = 235\text{Gb} + 1147\text{Gb} = 1382\text{Gb}$. In the consumption data sheet (Appendix A1), $c_d(2003) = 28.51\text{Gb}$. $r_d(2003)$ and $c_d(2003)$ are the initial data for the model.

The world population growth rate is 1.2% [United Nations 2004], and the present GDP growth rate is 2% (Martin 2004). We set the population growth rate $\mu = 0.012$ and the GDP growth rate $\lambda = 0.02$ in model 2, and the political factor equals to 1, we obtain that in year 2064, the remnant amount of oil is 25.62Gb, which is less than the current consumption amount of 27.83Gb. Therefore, in year 2064 oil will quit the energy market and we consider year 2064 as the depletion year of oil. This result is similar to the result of model 1 (the depletion year is 2062). It means our model is effective. Figure 6 shows the change of remnant oil with time. In figure 7, we can find that the oil consumption per year increases between year 2004 and 2017 and then decrease. The former growing is caused by the increase of economy and the population, however, as the finite remnant oil decrease, the annual consumption falls down afterward.

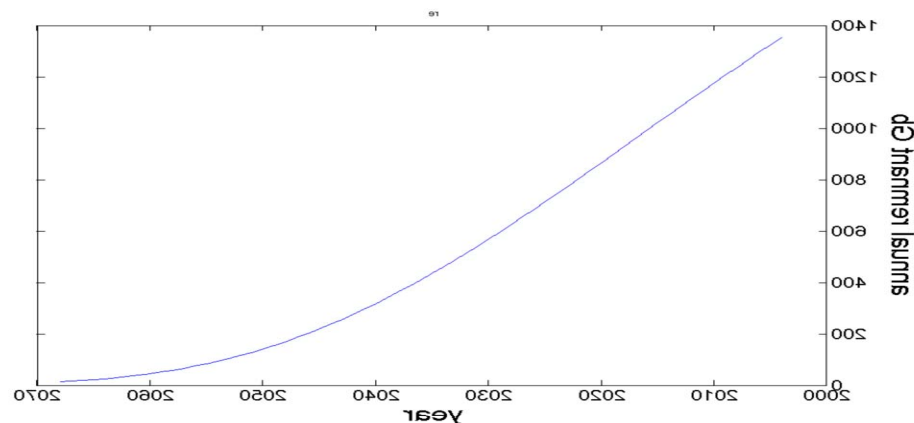


Figure 6 The Annual Remnant Oil at Current Growth Rate of GDP & population

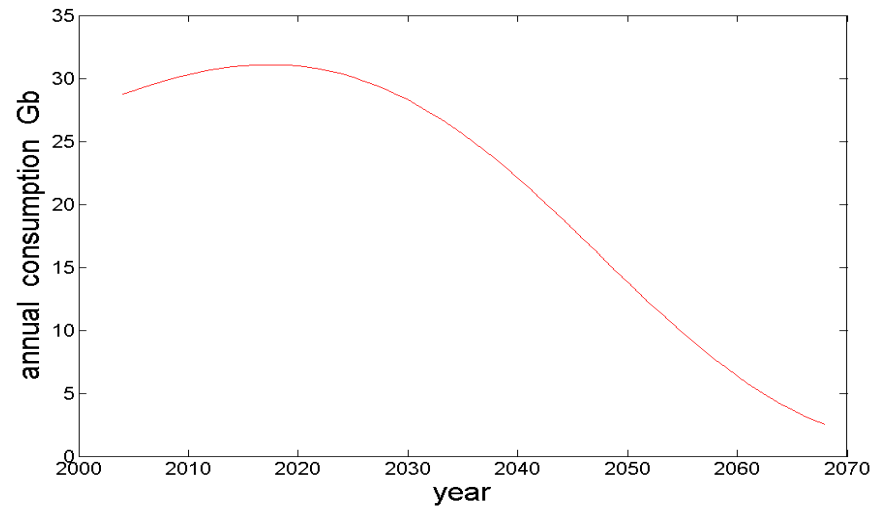


Figure 7 The Annual Consumption at Current Growth Rate of GDP & population

We can change the political factor, GDP growth rate, or population growth rate in the model to show the different effects of these factors on the depletion year of oil.

First, we try to find the impact of economy on the oil depletion. Because the world GDP growth rate is usually between 1% and 3%, we change the GDP growth rate to 0.01 and 0.03 separately, and the other two factors do not change in the model. For the GDP growth rate at 3%, the depletion year is 2055 that is 9 years earlier than the result for 2% GDP growth. For the GDP growth rate at 1%, the depletion year is 2078 that is 14 years later than the result for 2% GDP growth. The annual remnant oil and annual production for the GDP growth rate 1%, 2% and 3% is shown in Figure 8 and Figure 9.

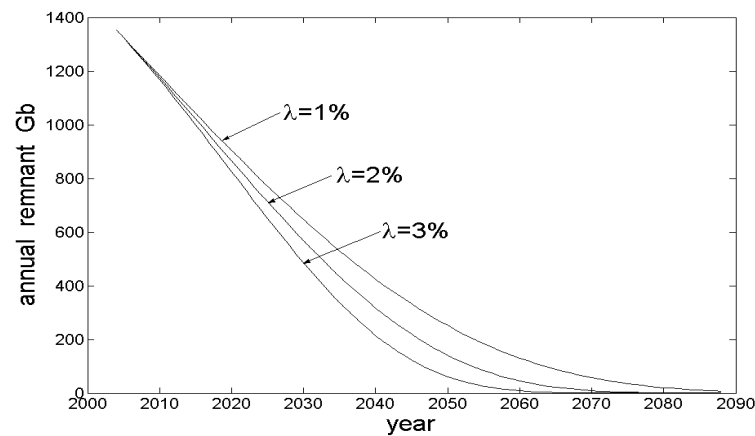


Figure 8 Annual Remnant Oil for Different GDP Growth Rate

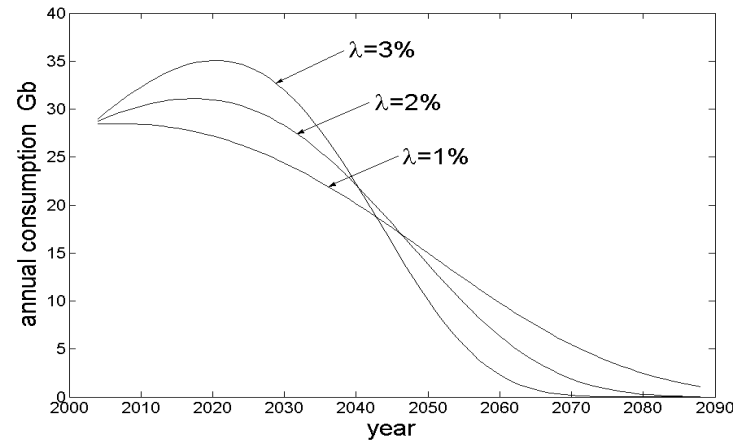


Figure 9 Annual Consumption for Different GDP Growth Rate

Second, we try to find the impact of population to the oil depletion. We change the population growth rate to 0.6% and 1.8% separately, and the other two factors do not change in the model. For the population growth rate at 1.8%, the depletion year is 2058 that is 6 years earlier than the result for 1.2% GDP growth. For the GDP growth rate at 0.6%, the depletion year is 2071 that is 7 years later than the result for 1.2% population growth. The annual remnant oil and annual production for the population growth rate 0.6%, 1.2% and 1.8% is shown in Figure 10 and Figure 11.

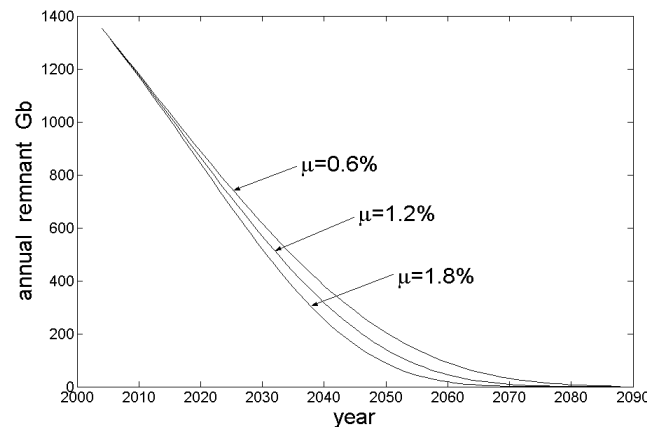


Figure 10 Annual Remnant Oil for Different Population Growth Rate

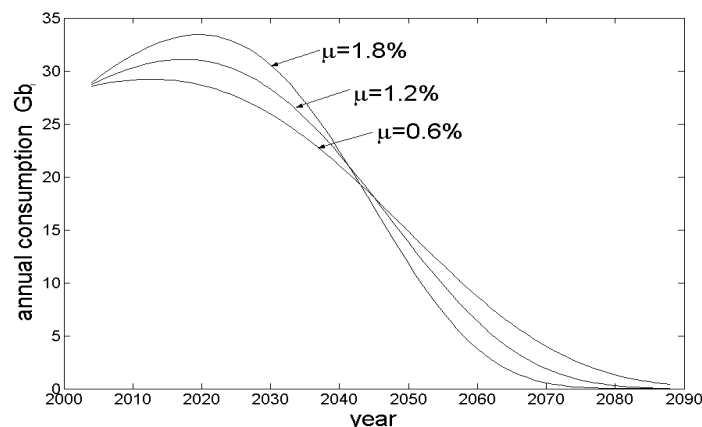


Figure 11 Annual Consumption for Different Population Growth Rate

Finally, we try to find the impact of political factor to the oil depletion. We assume that there is a political event that impedes the oil consumption for 10 years. Because the political event cannot be put into effect or lose effect abruptly, we set a series of political factors which are larger than 1 and change gradually for 10 years. The annual remnant oil and annual production for this political event and no political event is shown in Figure 12 and Figure 13.

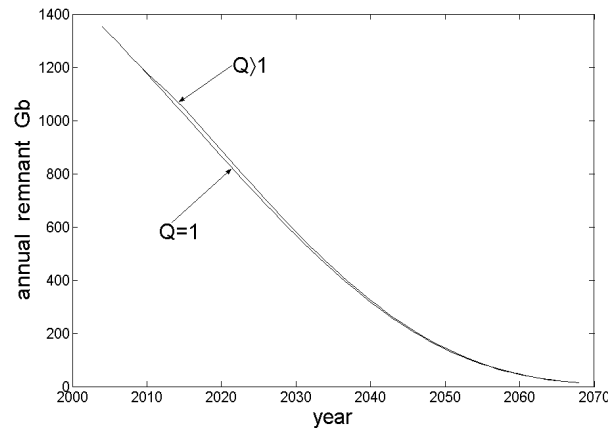


Figure 12 Annual Remnant Oil for Different Political Factor

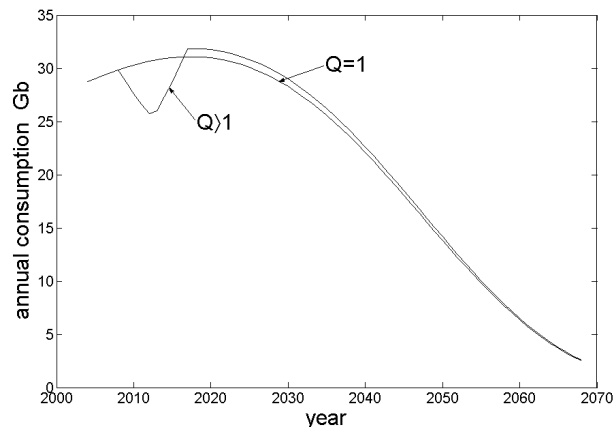


Figure 13 Annual Consumption for Different Political Factor

In conclusion, the GDP growth rate and the population growth rate both have significant effect on the oil depletion year and the consumption, however, the political event only affect the consumption significantly and has little impact on the oil depletion year.

Model 3 Prediction for the oil price

Because oil is running out year by year, the price of the oil will surely increase. A little change of the oil price will affect the world economy much, so to predict the world oil price is a valuable task.

According to the economics theory, the price of oil decrease with the supply increase, and increase with demand increase. To simplify the problem, we assume the supply, demand, and price meet the following equation:

$$\frac{\text{supply}}{\text{demand}} \times \text{price} = \text{constant}$$

With the consumption data from 1986 to 2003 (Appendix A1) we could figure out the constant is 25.4436. Because the year 2003 is near the peak production, and the supply cannot completely meet the

demand, we can consider the supply equals to the consumption from 2003. Therefore, we use the predicting data of consumption in model 1 as the supply in future. The supply is shown in Figure 15.

We assume the demand growth rate is the same as GDP growth rate. The oil demand for the world in 2004 is 80,000 thousand barrels a day, and it is 29.2Gb a year[<http://www.cniti.org/newsview.asp>]. Therefore:

$$\text{demand} = 29.2 \times (1 + \lambda)^{t-2004}$$

where λ is growth rate of GDP

We assume λ equals to the current world GDP growth rate 2%. Then we can calculate the oil price of each year. The prediction price of each year is shown in Figure 14.

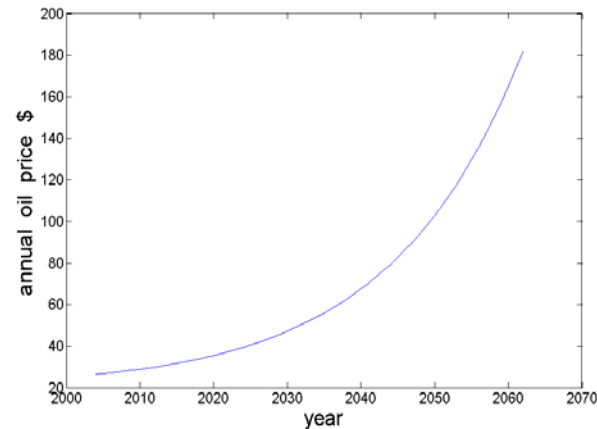


Figure 14 Future Oil Price

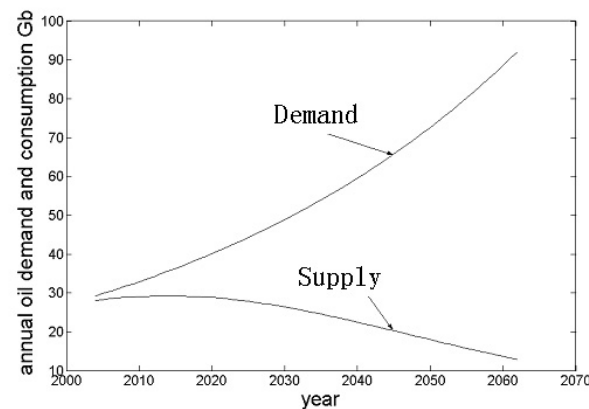


Figure 15 Future Oil Demand & Supply

In Figure 14, we can find that the oil price will rise to about 180\$ per barrel in the oil depletion year 2062.

Although the supply of oil first grows during the year 2004-2013 showed in Figure 2, the price is increasing incessantly in Figure 14. That is because the demand increases faster than the supply. It is shown in Figure 15.

Strength and limitation for the model

Strength

The Hubbert curve is the primary analytical tool for understanding and predicting the production of oil,

so the result of model 1 is reliable at a certain extent.

Model 2 includes the parameters of economic, demographic and political factors, so it is suitable for various condition of the society.

The parameters of model 2 is not fixed, it can be adjusted if the society factors changed.

The result of these models can be achieved efficiently by computer.

Limitation

We just use limited history data to fit the consumption and discovery in model 1, so the curve may not be absolutely accurate.

We do not take into account the deepwater exploration's effect on the discovery of oil in future.

The model 2 is based on the result of model 1. It cannot rectify the deviation of model 1. In model 2, the value of political factor is hard to determine.

The oil security

Oil is very important for the economy development of every country. It has turned into a strategy resource. We have to take some measures to prevent the oil from risks such as theft, misuse, disruption, and unnecessary degradation of the oil. In all the risks, how to prevent the oil disruption is an emergent problem.

Nowadays, the oil export country is mainly in the area of Gulf. Once some accident like wars occurs in the Gulf Area, the total production of the oil will surely decrease abruptly, and then the country depending on the oil import from Gulf will suffer a lot of loss for the oil deficiency. Therefore, to build a Strategic Petroleum Reserve (SPR) can protect the oil security for the country. In the following, we discuss how to decide the optimal levels of SPR.

We use a national perspective to analyze the problem. This analysis is based on the assessment of the costs and benefits for SPR. We measure the benefits of SPR as the avoided costs of a supply disruption; the costs are estimated from the costs of product and of storage. Thus, we are effectively seeking the optimal level of SPR that can achieve the most absolute benefits that exclude the cost.

Cost of SPR

The benefits of storage calculated in the preceding section need to be balanced against storage costs, which broadly comprise three elements:

1. The cost of constructing tanks
2. The cost of crude or product to fill the tanks
3. Cost of operating and maintaining the tanks and the product inventory

Assuming the price for each tank is p_t , the volume of each tank is v_t , the total storage is v , we get the total cost of construction tanks c_t :

$$c_t = \frac{v}{v_t} p_t$$

Assuming the price for the crude oil is p_o , the total cost of crude to fill the tanks is c_o , we get:

$$c_o = vp_o$$

The cost for operating and maintaining the tanks and the inventory is decided by the number of workers in need and the salary for each worker. The more oil is stored, the more workers are needed.

Assuming the relation between the number of workers and the volume of storage is direct ratio, whose ratio factor is k , we get the operating and maintenance costs c_m by the equation

$$c_m = kvs$$

where

v is the total storage;

s is the salary for each worker.

The total cost for storage is the sum of the tank construction costs, product costs, and operating and maintenance costs:

$$\text{Total Cost} = c_t + c_o + c_m$$

We find each of three parts (c_t , c_o , c_m) is the direct ratio of the storage volume v , so the total cost for storage is the direct ratio to the volume of the storage. The relation is showed in Figure 16:

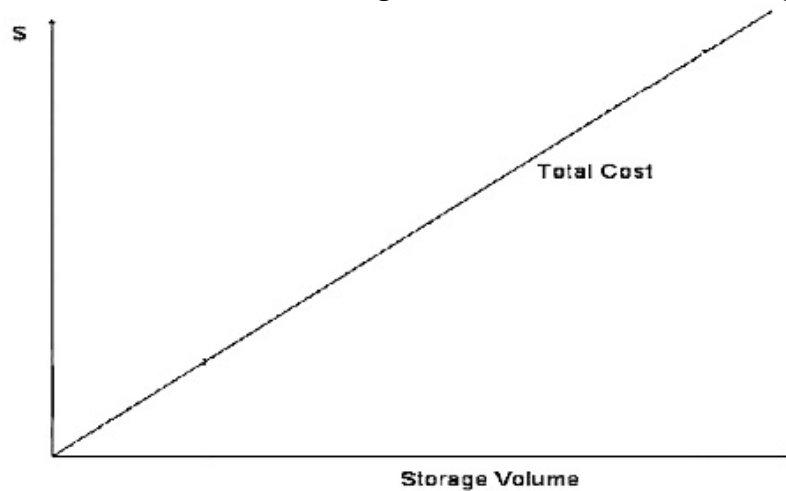


Figure 16 The Total Cost of SPR

Benefits of SPR

The benefits of storage come from saving the loss by maintaining the supply during the disruption. The SPR can completely meet the oil demand that the total shortfall in the disruption is less than or equals to the storage level and can partly meet oil demand that the total shortfall in the disruption is larger than the storage level. For example, a 5 days SPR can completely meet the oil demand during the disruption which lasts less than 6 days, and only can meet 5 days requirement during the disruption last more than 5 days.

Assuming the oil requirement for each day is r , the loss caused by oil disruption per day is l , the probability of occurring n days disruption is P_n , the total storage is v , the days that the storage can sustain is $m = v/r$, the probability of occurring more than m days disruption is p , the benefits of storage is b , and the benefits equals to the loss that the storage can prevent, we get:

$$b = \sum_{n=1}^m (P_n nl) + Pml$$

We know that as the volume of the storage increases, the increasing rate of the benefits of storage decreases. The approximate relation between the benefits and the volume of the storage is shown in Figure 17

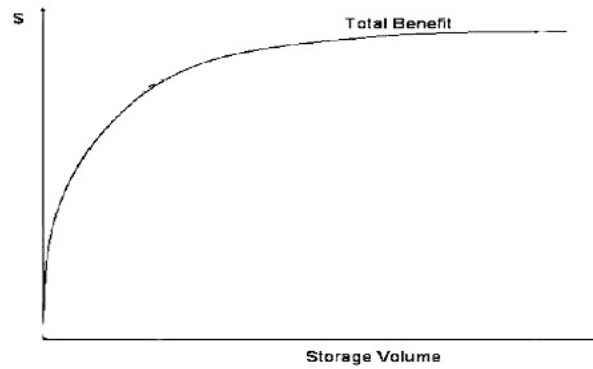


Figure 17 The Total Benefit of SPR

Assessment concept for optimal level of SPR

To find the optimal level of SPR, we must take into account both the benefits and cost. We can use the economics theory to solve this problem. We define the **marginal benefits** as the slope of the total benefit line, and **marginal cost** as the slope of the total cost line. We combine the Figure 16 and Figure 17 as follow:

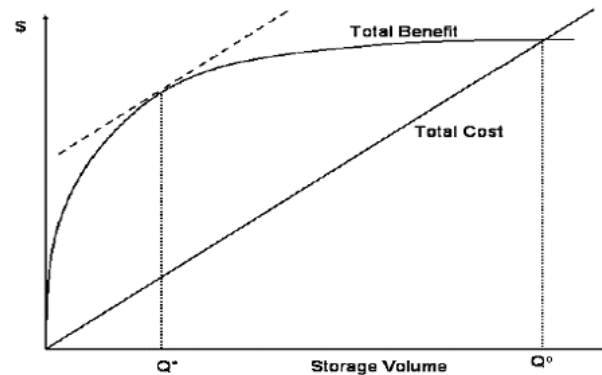


Figure 18 Total Cost and Benefit

Different country can have different criterion to find the optimal level of SPR. For the country that is not wealthy enough, the criterion to maximize the total retained benefit (benefit-cost) may be a good choice. For the country that has enough money, it can adopt the criterion that there is no retained loss (benefit-cost ≥ 0) for the storage.

If the criterion for socially optimal storage is to maximize the total retained benefit, (see Figure 18), this corresponds to finding the level of storage at which the gap between benefit and costs is largest, which is at the point Q^* . If we had storage of Q^* , there would be no incentive to store more, because the marginal benefit of further storage (i.e. points to the right of Q^*) are less than the marginal costs of storage. Picking the level of storage at which the marginal benefits and marginal costs of storage are equal will create the largest possible retained benefit from storage.

If the criterion is to maximize the storage with no retained loss, in Figure 18 we can find the point Q_0 , which is where the total costs and benefit of storage are equalized, is the optimal level of SPR.

How to sustain the oil usage

The production of the oil is approaching the peak oil day after day, however, the demand for the oil is increasing incessantly. The supply of the oil cannot meet the great demand. Hence, we need to take a proper policy to incen the oil production capability and control the increasing demand so as to sustain the oil usage for a long period..

The policy for increase the oil production is deepwater exploration. Deepwater exploration can

contribute a high increase to the world oil production. However, there are three major technological challenges for the deepwater exploration: the ability to drill cost effectively, deeper reservoirs with higher pressure and higher temperatures, and the need to better understand how reservoirs perform in these extreme water depths and subsea depths and under the pressure-temperature conditions encountered. We should try to improve the technology to solve all these problems, and then deepwater exploration can do more contribution to sustain the oil usage.

There are many kinds of measures to decrease the consumption of the oil. First, we can update and enforce building codes so that buildings will require much less energy while having very good indoor air quality. Second, increase the use of solar and wind energy before we run short of the materials and energy needed to produce them. Third, stop using tax dollars to subsidize the fossil fuel industry (let fossil fuels compete in a free market against energy efficiency and renewable energy sources). Another practical approach would be the electrification of transport. Switching half the truck and personal auto miles to electrified transport would require an increase in electric generation capacity of only 10 percent. Electrified transport has a lot of merits such as clean, non-polluting and energy-efficient.

Policy for Controlling the Pollution of Oil

Millions of tons of oil pollute the world's oceans every year. It leaks from boats, and washes off streets during rains. Sometimes a tanker carrying oil runs into bad luck and spills its cargo directly in the sea. Oil pollutes lakes and streams as well, when it washes in from streets during rains and seeps from boats.

Cleaning up an oil spill is nearly impossible. Scientists are working hard to figure out ways to respond quickly to oil spills and to reduce the damage spills do to the environment. So the proper storage of oil is very important.

Oil must be stored in a container of sufficient strength and structural integrity to ensure that in normal circumstances it is unlikely to leak or burst. This means that the container is expected to last until replacement, with proper maintenance, without causing, or being at risk of causing, pollution. It is recommended that you purchase a fixed container expected to last for a minimum of 20 years before it needs to be replaced. Note, plastic tanks are not suitable for the storage of flammable liquids. When in use, containers must at all times meet the performance standards laid down.

It is recommended that, where practicable, primary containers (whether tanks, intermediate bulk containers, mobile bowsters or drums) should not be constructed or situated outside a building within 50 meters of any borehole or 10 meters of any inland freshwaters and coastal waters that any leaking oil could enter. This includes rivers, lakes, reservoirs and smaller watercourses such as streams and ditches as well as perforated drainage pipes. It is recognized that this is not always practicable, for example, where primary containers are located in boat yards or in coastal locations. However, in these cases, or where it is difficult to fit a bund, it is important to seek advice from the Agency. Routine maintenance of the primary and secondary containment systems to prevent any risk of water pollution is very important. It means a more detailed annual inspection and service. Maintenance proposals should not be onerous as storage tanks have few mechanical features.

A secondary containment system is a further container to catch any oil leaking from the primary container or its ancillary pipework and equipment. This may take the form of a 'bund' used for primary containers, including multiple drums or a drum storage area, or a 'drip tray' for a mobile bowster or drum or any other containment system for preventing oil that is no longer in the container from escaping from the place where it is stored.

Transportation of the oil is another problem. Tankers to install segregated ballast tanks (SBTs), which remove a major source of oil pollution from ships is a very good way. Tanker owners installed SBTs by required dates, even though this entailed significant investments with no offsetting benefits and even though decreasing oil prices were increasing the pressure to cut costs.

Every vessel shall be provided with an oil record book in which every operation connected with the transfer of oil from or to the vessel shall be recorded. The oil record book shall be kept by the master of the vessel in the form prescribed by regulations. The Minister of the Interior shall appoint inspectors for

the prevention of pollution of the sea by oil. Inspector may enter a vessel and any other place containing installations or materials likely to pollute seawater by oil. They could take samples of any liquid that in his opinion is likely to be a source of pollution of the sea by oil. So the exact information about the vessel could be confirm. Then we can decide how to deal with the vessels. Strict discharge standards should be establish to limit the amount and location of discharges. If any oil is discharged, or allowed to escape, whether directly or indirectly, into seawater from any vessel or from any place on land or from any apparatus, as the case may be, shall be liable to a fine.

Alternative to oil

Oil is running out. According to our prediction, the oil can only be used for about 60 years. It is essential for people to find alternative to oil. There are many kinds of alternatives that can replace the purpose of the oil. The alternatives to oil include gas, coal, and nuclear power. They all have some advantages and disadvantages compared to the oil.

Gas

Advantage

Gas has many benefits to the power companies, producing more energy per tonne than oil and coal, and fewer pollutants such as carbon dioxide. It is used to create nitrogen fertilizers for agriculture. It will last longer than oil

Disadvantage

Gas, being a hydrocarbon, suffers from the same problem of depletion as oil. Otherwise, the main disadvantage is that it is not as easy to transport, requiring either pressure pipelines or special ships which pre-process it into LNG (liquefied natural gas). Both of these are at great risk to terrorism (a fully loaded LNG ship is apparently equal to 55 Hiroshima nuclear bombs). Gas is best used for electricity production and heating, so cannot supply the fuels and products that oil is used for. Conversion to a fuel suitable for aircraft and ships uses large amounts of energy and money.

Coal

Advantage

Coal is far more abundant than oil and gas. There is about 200 years of coal left (but only 100 years if it was used to replace oil and gas). Coal does not peak like oil. The largest reserves are often in areas that are low in oil and gas, thereby reducing the risk of resource wars. Coal can be converted into oil and gas.

Disadvantage

Coal is far more polluted than oil and gas, producing ash and flue gasses which contain sulphur dioxide, nitrogen oxides, arsenic among others. Its extraction can create subsidence, spoil heaps and strip mining damage, as well as many deaths among miners. It is less convenient to store and use, and it produces up to twice as much carbon dioxide. Something like 50% of the energy used to mine coal comes from oil. It is also more difficult to transport and control than oil so usage is expensive and wasteful.

Nuclear Power

Advantage

Despite claims to be carbon dioxide free, energy generation from nuclear power *does* contribute to global warming, although not as much as oil, gas and certainly coal. Unlike renewable energies, the technology and structures for nuclear power are already in place so, unless we wanted to build more, much of the cost and work has already been done.

Disadvantage

Nuclear electricity generation does have an awkward balance - keeping down carbon dioxide omissions would rapidly use up the stores of rich uranium, while using other grades of uranium would increase the amount of carbon dioxide emitted. To replace oil on its own, we would need to build

hundreds of new stations, each with the resulting cost and radioactive pollution. It takes many years and a great deal of energy to both build and decommission nuclear power plants. The public has already been scared off of nuclear power and the politicians would not risk returning to that source, certainly while the public is unaware of the oil depletion danger. Like most alternatives, nuclear power can only supply energy, not the multitude of other products that oil gives us.

Policy

Even though the alternatives above can replace the purpose of oil in some way, these alternatives are all nonrenewable resource, and they will be run out at last. In the long term, renewable energy sources are the only hope for mankind. They are the only forms that will not run out and are therefore less likely to lead to resource wars. There are many kinds of renewable energy: Biomass, Hydroelectricity, Wind Power, Sea Power, Geothermal, and Solar Energy. The costs of manufacture and the length of time needed to fully develop are major obstacles for the wide use of these forms of renewable energy. In future, each country need to put into a lot of money to do some research on how to economically make use of these renewable resource and how to enlarge the applied field for these forms of energy. The country that makes widely use of the renewable energy can have a nicety future.

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Appendix

All the petroleum data (A1 – A5) is from <http://www.bp.com/statisticalreview2004>

A1

Table 1 World Annual Petroleum Consumption

year	1965	1966	1967	1968	1969	1970	1971	1972	1973
Gb	11.40	12.28	13.17	14.25	15.51	16.82	17.75	19.04	20.57
	1973	1974	1975	1976	1977	1978	1979	1980	1981
	20.27	20.06	21.34	22.13	23.04	23.46	22.54	21.83	20.27
	1982	1983	1984	1985	1986	1987	1988	1989	1990
	21.21	21.13	21.57	21.55	22.21	22.67	23.32	23.77	24.17
	1991	1992	1993	1994	1995	1996	1997	1998	1999
	24.19	24.42	24.34	24.85	25.27	25.82	26.50	26.65	27.24
	2000	2001	2002	2003					
	27.55	27.71	27.97	28.51					

A2

Table 2 World Annual Petroleum Production

year	1965	1966	1967	1968	1969	1970	1971	1972	1973
Gb	11.61	12.62	13.55	14.76	15.93	17.54	18.56	19.59	21.34
	1973	1974	1975	1976	1977	1978	1979	1980	1981
	21.40	20.38	22.05	22.89	23.12	24.11	22.98	21.73	21.40
	1982	1983	1984	1985	1986	1987	1988	1989	1990
	20.91	20.66	21.05	20.98	22.07	22.18	23.04	23.36	23.88
	1991	1992	1993	1994	1995	1996	1997	1998	1999
	23.80	23.99	24.09	24.47	24.82	25.48	26.29	26.79	26.30
	2000	2001	2002	2003					
	27.25	27.19	27.03	28.02					

A3

Table 3 World Petroleum Reserves

year	1980	1981	1982	1983	1984	1985	1986	1987	1988
Gb	669.6	683.0	712.1	723.0	763.3	772.0	883.2	913.2	1002.2
	1989	1990	1991	1992	1993	1994	1995	1996	1997
	1020.2	1015.9	1019.4	1023.0	1023.6	1031.1	1040.5	1063.8	1053.1
	1998	1999	2000	2001	2002	2003			
	1066.3	1083.5	1106.1	1114.3	1146.3	1147.7			

A4**Table 4 Annual Petroleum Discoveries**

year	1980	1981	1982	1983	1984	1985	1986	1987	1988
Gb	11.61	12.62	13.55	14.76	15.93	17.54	18.56	19.59	21.34
	1989	1990	1991	1992	1993	1994	1995	1996	1997
	21.40	20.38	22.05	22.89	23.12	24.11	22.98	21.73	21.40
	1998	1999	2000	2001	2002				
	20.91	20.66	21.05	20.98	22.07				

A5**Table 5 Crude Oil price since 1985**

US dollars per barrel	Year	\$ money of the day	\$ 2003
	1985	27.53	47.18
	1986	14.32	24.14
	1987	18.33	29.83
	1988	14.92	23.27
	1989	18.23	27.02
	1990	23.73	33.54
	1991	20.00	27.11
	1992	19.32	25.41
	1993	16.97	21.74
	1994	15.82	19.84
	1995	17.02	20.75
	1996	20.67	24.43
	1997	19.09	22.14
	1998	12.72	14.80
	1999	17.97	20.16
	2000	28.50	30.96
	2001	24.44	25.75
	2002	25.02	25.77
	2003	28.83	28.83

B1**Program 1 Nonlinear Fitting on Annual Consumption**

```

clear
tdata=[1985:2003];
cdata=[21.55 22.21 22.67 23.328 23.77 24.17 24.19 24.42 24.34 24.86 25.27 25.82
26.50 26.65 27.24 27.55 27.71 27.97 28.51];
x0=[40.8 0.01 2000];
[x,resnorm]=lsqcurvefit('hubbert',x0,tdata,cdata)

```

The m-file of Hubbert function is as follow:

```
function c=hubbert(x,tdata)
```

```
c=2*x(1)./(1+cosh(x(2)*(tdata-x(3))));
```

B2

Program 2 Nonlinear Fitting on Annual Discoveries

```
clear
tdata=[1980:1984,1986,1988:1990];
ddata=[36.42 50.87 31.79 60.93 29.78 52.02 41.03 18.98 24.67];
x0=[60 0.01 1960];
[x,resnorm]=lsqcurvefit('hubbert',x0,tdata,ddata)
```

The m-file of Hubbert function is as follow:

```
function c=hubbert(x,tdata)
c=2*x(1)./(1+cosh(x(2)*(tdata-x(3))));
```

B3

Program 3 Calculate the Total Remnant oil

```
clear
sy=1382;
csm=27.829;
k=csm/sy;
for n=1:65
    csm=k*((1+0.02)*(1+0.012)).^n*sy;
    sy=sy-csm;
    s(n)=csm;
    syu(n)=sy;
end
i=2004:2068;
figure
plot(i,s,'r-');xlabel('year');ylabel('annual consumption Gb');
figure
plot(i,syu,'b-');xlabel('year');ylabel('annual remnant Gb');
```

B4

Program 4 Calculate the oil price

```
n=2004:2062;
d=29.2*(1+0.02).^(n-2004);
c=[28.065 28.28 28.474 28.647 28.798 28.927 29.033 29.115 29.174 29.21 29.222 29.21 29.174 29.115
    29.033 28.927 28.798 28.647 28.474 28.28 28.065 27.829 27.575 27.301 27.01 26.702 26.378 26.039
    25.685 25.318 24.939 24.548 24.147 23.736 23.317 22.891 22.458 22.019 21.576 21.129 20.679
    20.227 19.774 19.321 18.867 18.415 17.964 17.515 17.07 16.627 16.189 15.755 15.326 14.903
    14.485 14.073 13.667 13.268 12.876];
size(c),
bili=c./d;
p=25.4436./bili;
figure
plot(n,p);xlabel('year');ylabel('annual oil price $ ');
figure
plot(n,d);xlabel('year');ylabel('annual oil demand Gb');
```

calculate the constant l

```
clear
p=[24.14 29.83 23.27 27.02 33.54 27.11 25.41 21.74 19.84 20.75 24.43 22.14 14.80 20.16 30.96 25.75 25.77 28.83];
n=1986:2003;
d=29.2*(1+0.02).^(n-2004);
s=[22.06 22.17 23.03 23.35 23.87 23.79 23.98 24.09 24.47 24.82 25.47 26.28
26.79 26.30 27.25 27.18 27.03 28.02];
bili=s./d,
x0=[20];
[x,resnorm]=lsqcurvefit('pri',x0,bili,p)
```

The m-file of function pri is as follow:

```
function y=pri(x,xdata)
y=x(1)./xdata;
```